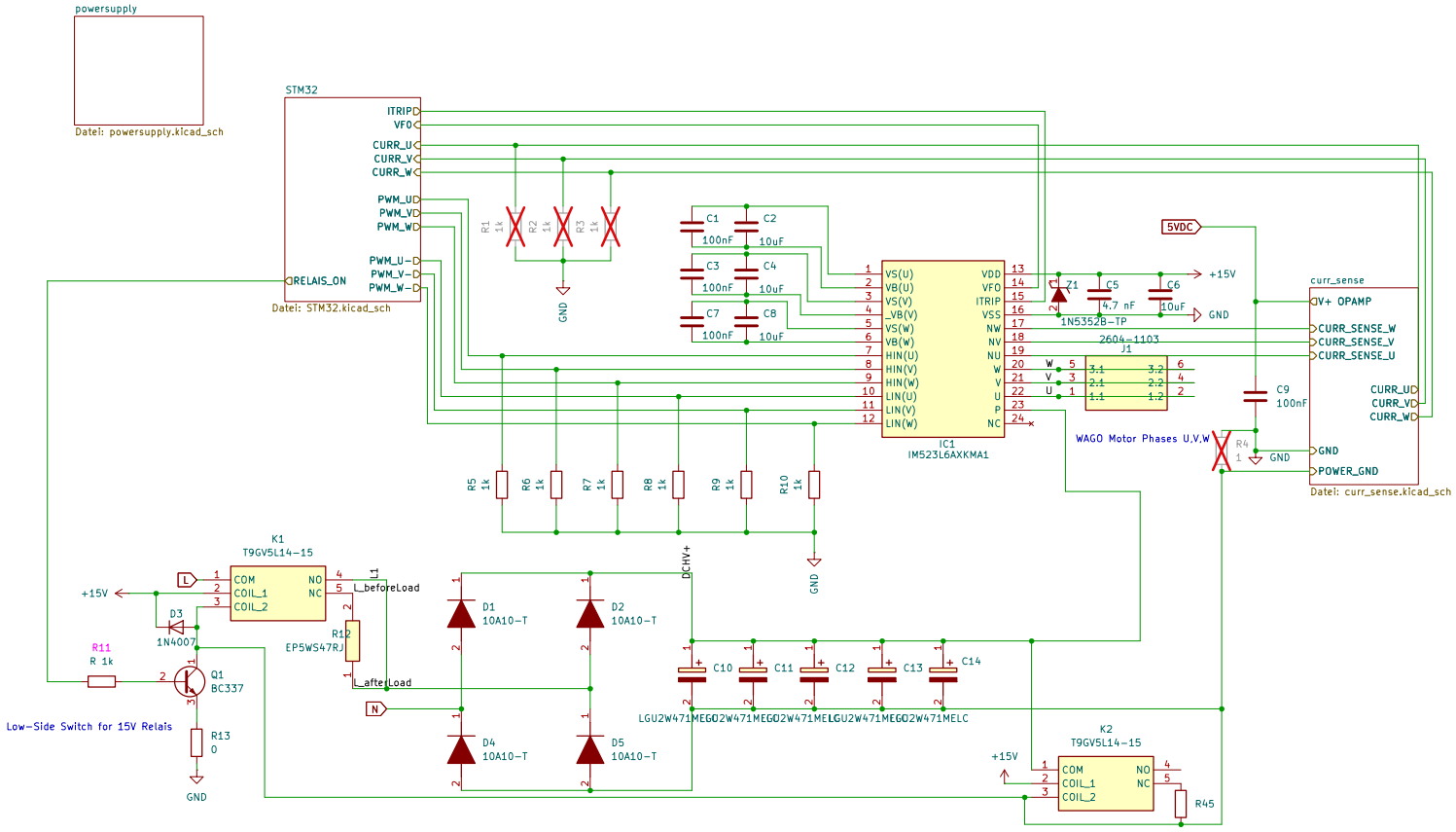
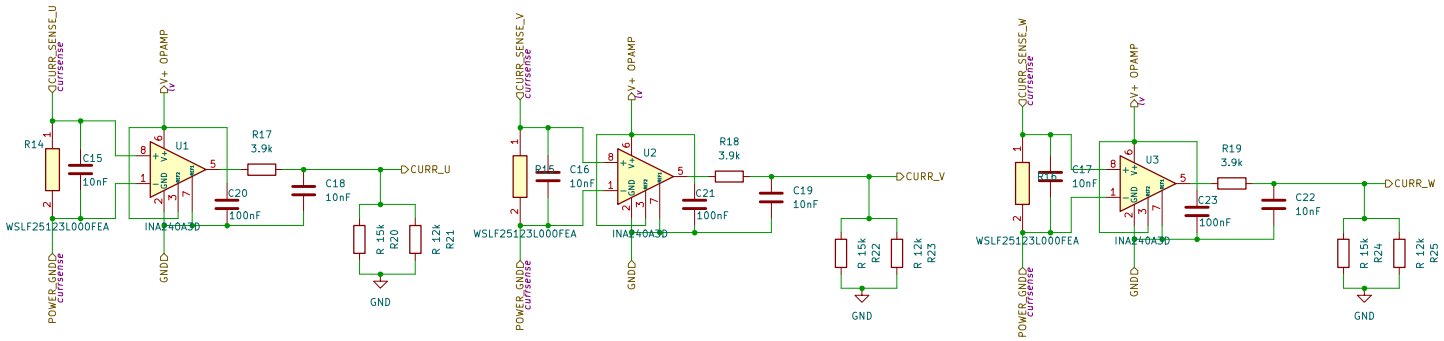
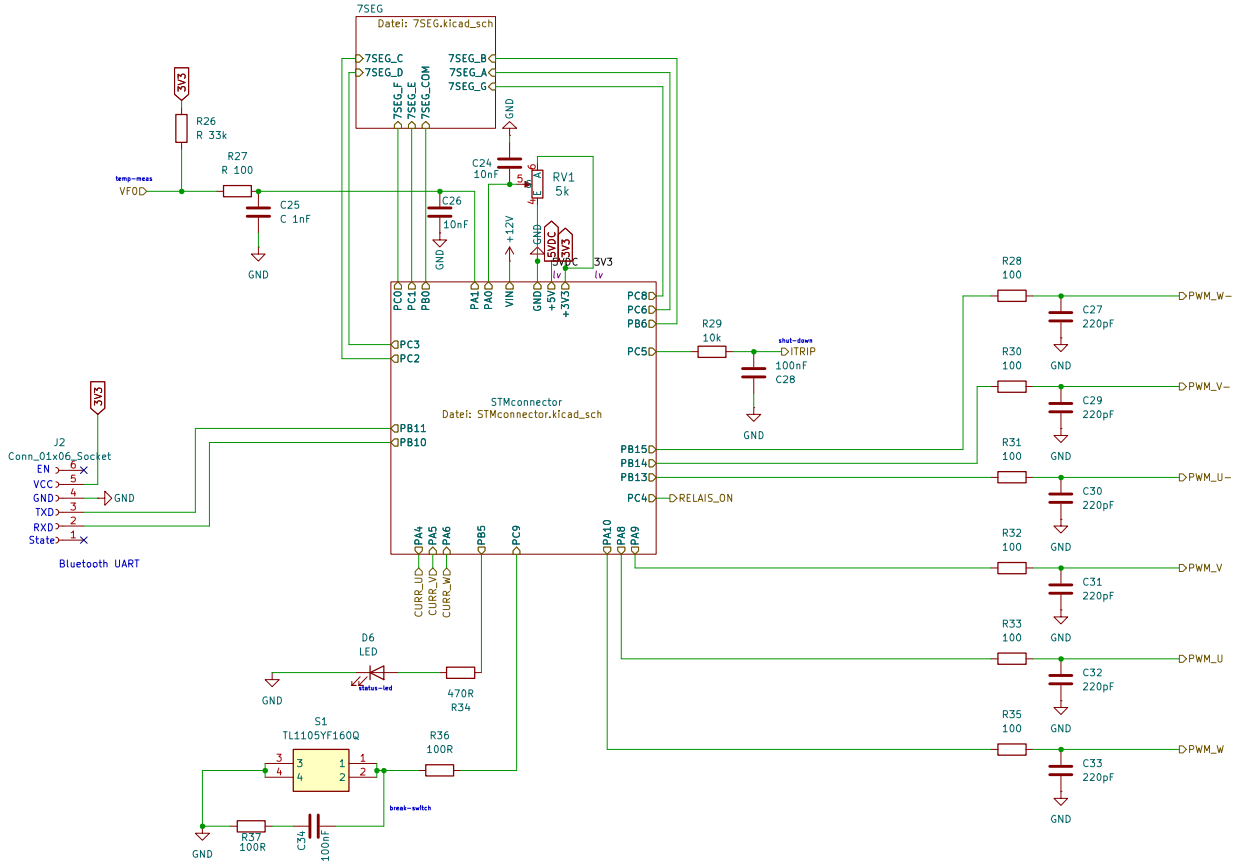
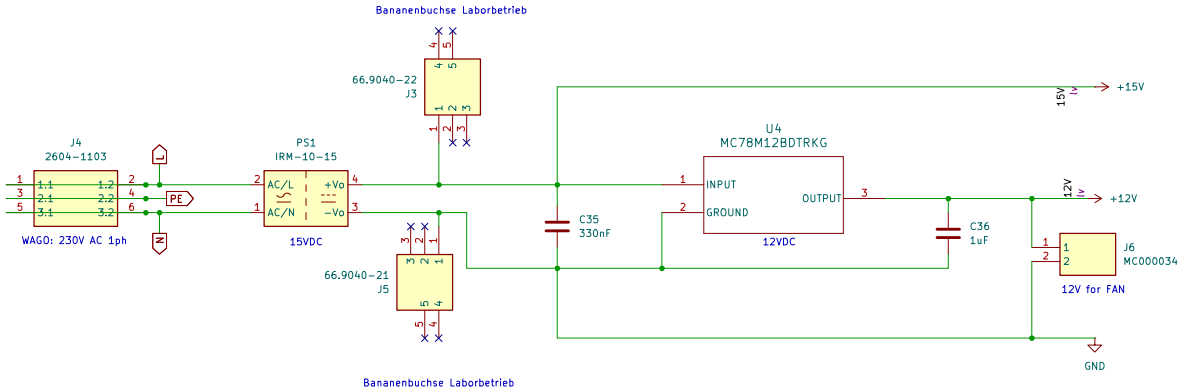
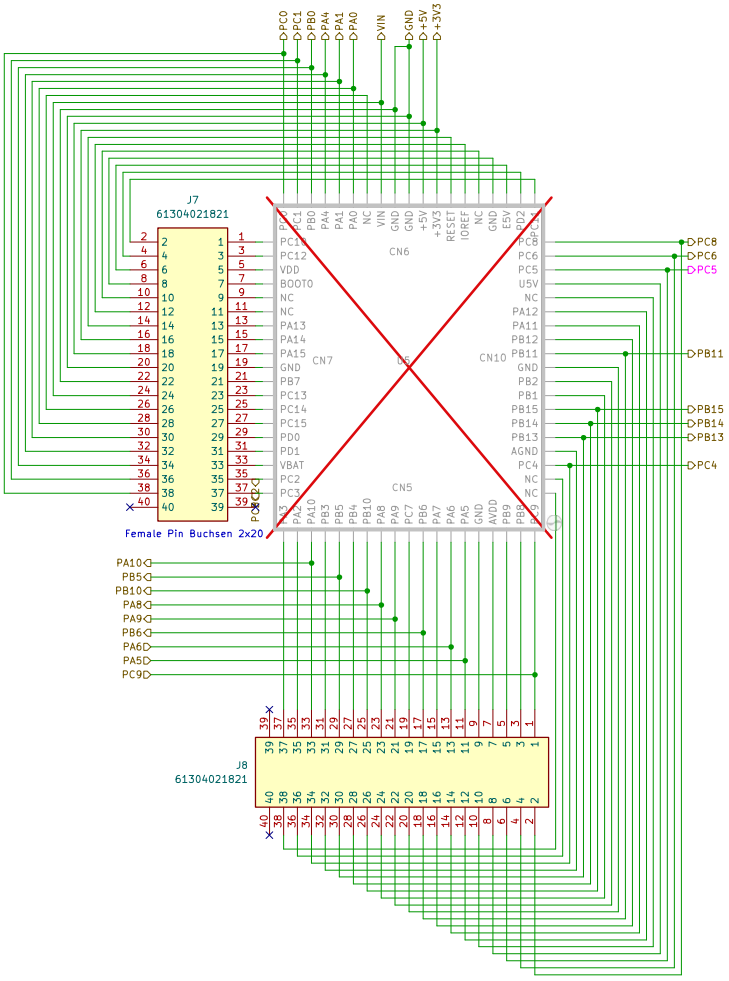

powersupply
Datei: powersupply.kicad_sch
STM32
ITRIP
VFO
CURR_U
CURR_V
CURR_W
PWM_U
PWM_V
PWM_W
RELAIS_ON
Datei: STM32.kicad_sch
R1 1k
R2 1k
R3 1k
GND
C1 100nF
C2 10uF
C3 100nF
C4 10uF
C7 100nF
C8 10uF
IC1
IM523L6AXKMA1
VS(U)
VB(U)
VS(V)
_VB(V)
VS(W)
VB(W)
HIN(U)
HIN(V)
HIN(W)
LIN(U)
LIN(V)
LIN(W)
VDD
VFO
ITRIP
VSS
NW
NV
NU
W
V
U
P
NC
5VDC
+15V
Z1
1N5352B-TP
C5 4.7 nF
C6 10uF
GND
J1
2604-1103
WAGO Motor Phases U,V,W
C9 100nF
R4
GND
curr_sense
V+ OPAMP
CURR_SENSE_W
CURR_SENSE_V
CURR_SENSE_U
CURR_UD
CURR_VD
CURR_WD
GND
POWER_GND
Datei: curr_sense.kicad_sch
R5 1k
R6 1k
R7 1k
R8 1k
R9 1k
R10 1k
GND
K1
T9GV5L14-15
COM
COIL_1
COIL_2
NO
NC
L
+15V
D3
1N4007
R11
R 1k
Q1
BC337
R13
0
GND
Low-Side Switch for 15V Relais
L1
L_beforeLoad
R12
EP5WS47RJ
L_afterLoad
N
D1 10A10-T
D2 10A10-T
D4 10A10-T
D5 10A10-T
DCHV+
C10
C11
C12
C13
C14
LGU2W471MEGO2W471MEGO2W471MELGU2W471MEGO2W471MELC
K2
T9GV5L14-15
+15V
R45

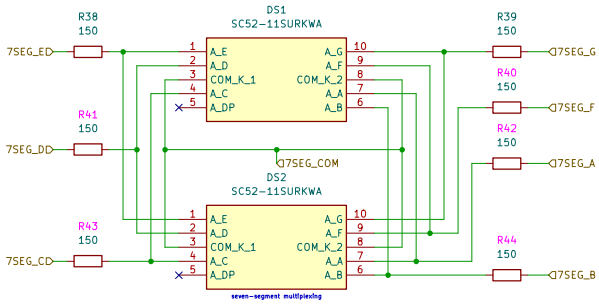

CURR_SENSE_U
currsense
5V+ OPAMP
5v
R14
C15
10nF
U1
R17
3.9k
C18
10nF
CURR_U
C20
100nF
WSLF25123L000FEA
INA240A3D
GND
R 15k
R20
R 12k
R21
POWER_GND
currsense
CURR_SENSE_V
currsense
5V+ OPAMP
5v
R15
C16
10nF
U2
R18
3.9k
C19
10nF
CURR_V
C21
100nF
WSLF25123L000FEA
INA240A3D
GND
R 15k
R22
R 12k
R23
POWER_GND
currsense
CURR_SENSE_W
currsense
5V+ OPAMP
5v
R16
C17
10nF
U3
R19
3.9k
C22
10nF
CURR_W
C23
100nF
WSLF25123L000FEA
INA240A3D
GND
R 15k
R24
R 12k
R25
POWER_GND
currsense


7SEG
Datei: 7SEG.kicad_sch
7SEG_C
7SEG_D
7SEG_F
7SEG_E
7SEG_COM
7SEG_B
7SEG_A
7SEG_G
GND
3V3
R26
R 33k
R27
R 100
temp-meas
VFO
C25
C 1nF
C24
10nF
RV1
5k
C26
10nF
+12V
5VDC
3V3
PC0
PC1
PB0
PA1
PA0
VIN
GND
+5V
+3V3
PC8
PC6
PB6
PC5
PC3
PC2
PB11
PB10
PB15
PB14
PB13
PC4
R29
10k
shut-down
ITRIP
100nF
C28
RELAIS_ON
STMconnector
Datei: STMconnector.kicad_sch
J2
Conn_01x06_Socket
EN
VCC
GND
TXD
RXD
State
Bluetooth UART
PA4
PA5
PA6
PB5
PC9
PA10
PA8
PA9
CURR_UD
CURR_VD
CURR_WD
D6
LED
status-led
470R
R34
S1
TL1105SYF160Q
R36
100R
break-switch
R37
100R
C34
100nF
R28
100
C27
220pF
PWM_W-
R30
100
C29
220pF
PWM_V-
R31
100
C30
220pF
PWM_U-
R32
100
C31
220pF
PWM_V
R33
100
C32
220pF
PWM_U
R35
100
C33
220pF
PWM_W


Bananenbuchse Laborbetrieb
66.9040-22
J3
J4
2604-1103
WAGO: 230V AC 1ph
PE
PS1
IRM-10-15
AC/L
AC/N
+Vo
-Vo
15VDC
66.9040-21
J5
Bananenbuchse Laborbetrieb
U4
MC78M12BDTRKG
INPUT
GROUND
OUTPUT
12VDC
C35
330nF
C36
1uF
15V
+15V
12V
+12V
J6
MC000034
12V for FAN
GND



PC13 --> nucleo-onboard-button B1


R38
150
7SEG_ED
R41
150
7SEG_DD
R43
150
7SEG_CD
DS1
SC52-11SURKWA
A_E
A_D
COM_K_1
A_C
A_DP
A_G
A_F
COM_K_2
A_A
A_B
1
2
3
4
5
10
9
8
7
6
7SEG_COM
DS2
SC52-11SURKWA
R39
150
7SEG_G
R40
150
7SEG_F
R42
150
7SEG_A
R44
150
7SEG_B
seven-segment multiplexing